Guidelines for scientific writing

Helping beginners to avoid common mistakes

This document should help students to avoid common mistakes in their final theses. It contains guidelines on how to avoid common mistakes which we found in student theses. Please make sure that you follow these guidelines when writing your thesis.

1. Notation

Each variable and mathematical operator (except multiplication/addition) has to be unambiguously defined at the first time of usage. One common way is to use ...

- italic lower case characters for scalar variables (e.g. time index)
- italic upper case characters for scalar constants (e.g. filter length)
- boldface lower case characters for vectors (e.g. impulse response vector)
- boldface upper case characters for matrices (e.g. covariance matrix)
- same font types for same classes of variables (e.g., Greek letters for independent parameters)

Pay attention that the same notation is used in equations and figures!

2. Figures

Figures in a scientific report must be of high quality and not distorted, blurred or pixelated (in many cases redrawing with programs like Inkscape is the best solution to obtain a figure of good quality). Moreover, figures copied from or redrawn after some literature source have to be referenced (e.g. by a reference in the caption). Pay attention on appropriate axis labelling and consistent notation (see 1). Moreover, please refer to your figures from within the written text. A figure without reference or explanation in the text is useless. Finally, it is recommended that figures are placed at the top or bottom of a page, but still close to the parts of the text in which the figures are described.

3. Equations

Equations have to be interpreted as part of the text and may end with a punctuation character (e.g. with a comma or full-stop). Moreover, operators like “sin()“, “log()“, or “argmax()“ should not be italic (latex: e.g. \text{sin}). Only very compact mathematical expression should be written as in-line equation ($a=b$). For everything else, the equation environment should be used.

4. References

It is important to reference all figures, equations and fundamental statements which have been substantially adopted from different literature sources. Furthermore, check your list of references (at the end of the report) for consistency (e.g. abbreviations of authors should be equally formatted). Also take care that all your references are complete: References should at least contain the name(s) of the author(s), title of publication, type of publication (e.g., article or book), name of, e.g., journal in which the article was published , and the year of publication Please do not include references to Wikipedia!

5. In General

- Every statement in your thesis has to be justified, either by references or by yourself. For example, if you state that algorithm A has a better performance than algorithm B, cite the corresponding publication in which this was analysed or show a figure, where the two algorithms are compared with respect to their performance.
- Abbreviations (e.g. IR for impulse response) have to be introduced once and should be consistently used afterwards (e.g. using the latex-package “acronym“). If you introduce abbreviations, please add a list of abbreviations to your thesis.
- Avoid splitting equations which are defined the text (in the $$-environment) by using latex-operators like “\linebreak{}“ or “\mbox{}“.
- Introductory phrases like “In this section,“, “Furthermore,“ or “In Figure 2,“ always end with a comma.
- Do not create paragraphs which only contain one sentence.
- English grammar: Consider compound adjectives for dashes between adjectives: http://www.grammar.cl/english/compound-adjectives.htm
- Use a spell checker before giving the thesis to your supervisor!
- A good and efficient way to start writing your thesis is to start with an outline of your thesis. This enables you to structure your thesis on a very high level to avoid losing the overview while writing. In further steps, refine the chapters by lists of bullet points. These lists should contain the rough content of the respective chapter. You should be able to follow the main idea/results of your thesis based on this rough structure. Discuss the general structure of the thesis with your supervisor before starting to formulate complete sentences. After this, you can start filling each chapter with complete sentences and improve it iteratively.